LOCALIZATION AT A LEFT MULTIPLICATIVE SYSTEM COMMUTES WITH FINITE COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4654-4678, following the statement at 4647-4652.

Editorial note. The proof is complete with no omission markers; every equality of the chain is justified by a cited input or a universal property.

Allowed standard prerequisites (editorial).

- Hom formula for the localized category. equation-left-localization-morphisms-colimit, cited at proof line 4658. Supplies the first and fourth equalities of the chain at proof lines 4661-4671.
- Filteredness of Y/S and commutation of filtered colimits with finite limits. the left multiplicative system hypothesis and lemma-directed-commutes, cited at proof lines 4659-4660. Supplies the lim/colim exchange for the finite index category I at the third equality of the chain.

Source reference. Original equation 05Q1: sources/stacks-categories/pinned-original-source.tex, source lines 4614–4614. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 002W: sources/stacks-categories/pinned-original-source.tex, source lines 2399–2399. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category and let S be a left multiplicative system of morphisms of $\mathcal{C}$. The localization functor $Q : \mathcal{C} \to S^{-1}\mathcal{C}$ commutes with finite colimits.*

Proof. Let $\mathcal{I}$ be a finite category and let $\mathcal{I} \to \mathcal{C}$, $i \mapsto X_i$ be a functor whose colimit exists. Then using (05Q1), the fact that Y/S is filtered, and Lemma 002W we have

$$\begin{aligned}
\mathrm{Mor}_{S^{-1}\mathcal{C}}(Q(\mathrm{colim}\, X_i), Q(Y)) &= \mathrm{colim}_{(s:Y\to Y')\in Y/S} \mathrm{Mor}_{\mathcal{C}}(\mathrm{colim}\, X_i, Y') \\
&= \mathrm{colim}_{(s:Y\to Y')\in Y/S} \lim_i \mathrm{Mor}_{\mathcal{C}}(X_i, Y') \\
&= \lim_i \mathrm{colim}_{(s:Y\to Y')\in Y/S} \mathrm{Mor}_{\mathcal{C}}(X_i, Y') \\
&= \lim_i \mathrm{Mor}_{S^{-1}\mathcal{C}}(Q(X_i), Q(Y))
\end{aligned}$$

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

and this isomorphism commutes with the projections from both sides to the set $\mathrm{Mor}_{S^{-1}\mathcal{C}}(Q(X_j), Q(Y))$ for each $j \in \mathrm{Ob}(\mathcal{I})$. Thus, $Q(\mathrm{colim}\, X_i)$ satisfies the universal property for the colimit of the functor $i \mapsto Q(X_i)$; hence, it is this colimit, as desired. $\square$